

Week 08 — Spectral Estimation: Periodogram & Tapering

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The big picture. The periodogram estimates the spectral density but is biased (Fejér kernel) and inconsistent; tapering reduces bias, multitapering reduces variance.

We have a finite, equally spaced record $X_1, \dots, X_n$ ($T = \{1, \dots, n\}$, sampling interval $\Delta = 1$) and we want to estimate the spectral density function (SDF) $\mathcal{S}(f)$. The natural idea is a *plug-in*: since $\mathcal{S}(f) = \sum_{\tau} \gamma_{\tau} e^{-2\pi i f \tau}$, just replace γ_{τ} by the sample ACVS $\hat{\gamma}_{\tau}$. This gives the **periodogram**. The whole week is about what goes wrong with it (bias from a finite window, variance that never dies) and the two fixes (*tapering* for bias, *multitapering* for variance).

Key definitions

Definition — Periodogram

The plug-in estimator of the SDF: Fourier-transform the (biased) sample ACVS,

$$\hat{\mathcal{S}}^{(p)}(f) = \sum_{\tau \in \mathbb{Z}} \hat{\gamma}_{\tau} e^{-2\pi i f \tau}, \quad \hat{\gamma}_{\tau} = \frac{1}{n} \sum_{t=1}^{n-|\tau|} (X_t - \bar{X})(X_{t+|\tau|} - \bar{X})$$

for $|\tau| \leq n-1$ (and 0 otherwise). Equivalently it is the squared modulus of the finite Fourier transform of the centred data, $\hat{\mathcal{S}}^{(p)}(f) = |J(f)|^2$ with $J(f) = \sqrt{1/n} \sum_{t \in T} (X_t - \bar{X}) e^{-2\pi i t f}$.

Two faces of the periodogram

The lag form ($\sum_{\tau} \hat{\gamma}_{\tau} e^{-2\pi i f \tau}$) shows it is the honest plug-in for the SDF. The $|J(f)|^2$ form shows two things at a glance: it is **always** ≥ 0 (a squared modulus), and it can be computed by a single FFT in $O(n \log n)$. Same object, two viewpoints.

Definition — Data taper and tapered periodogram

A **data taper** is a window $\{h_t\}_{t \in T}$ normalised so that $\|h\|_2^2 = \sum_t h_t^2 = 1$. Replace the flat sum by a weighted one:

$$J_h(f) = \sum_{t \in T} h_t (X_t - \bar{X}) e^{-2\pi i t f}, \quad \hat{\mathcal{S}}_h^{(p)}(f) = |J_h(f)|^2.$$

The constant taper $h_t = \sqrt{1/n}$ recovers the ordinary periodogram. The normalisation $\|h\|_2^2 = 1$ is exactly what keeps the estimator unbiased for white noise.

Definition — Multitaper estimator

Take K **orthonormal** tapers, $\sum_{t \in T} h_{t,k} h_{t,k'} = \delta_{k,k'}$, and average their tapered periodograms:

$$\hat{\mathcal{S}}^{(mt)}(f) = \frac{1}{K} \sum_{k=1}^K \hat{\mathcal{S}}_{h_k}^{(p)}(f).$$

Its expectation is a convolution of $\mathcal{S}$ with the **average kernel** $\bar{H}(f) = \frac{1}{K} \sum_{k=1}^K |H_k(f)|^2$, and its variance is $\approx \mathcal{S}(f)^2 / K$.

Key results & formulas

Ideally an estimator would satisfy (1) $\mathbb{E}[\hat{\mathcal{S}}^{(p)}(f)] = \mathcal{S}(f)$, (2) $\text{Var}(\hat{\mathcal{S}}^{(p)}(f)) \rightarrow 0$, (3) Cov between distinct frequencies = 0. Reality: (1) holds only *approximately* (bias from the finite window), (2) is *false* (variance does not vanish), (3) holds approximately, but only at the Fourier frequencies.

Theorem — Expectation of the periodogram (Fejér kernel)

For a zero-mean process the expectation is a windowed ACVS, equivalently a frequency convolution with the Fejér kernel $\mathcal{F}_n$:

$$\begin{aligned}\mathbb{E}[\hat{\mathcal{S}}^{(p)}(f)] &= \sum_{\tau \in \mathbb{Z}} w_{\tau} \gamma_{\tau} e^{-2\pi i \tau f}, \quad w_{\tau} = \left(1 - \frac{|\tau|}{n}\right) \mathbf{1}_{[-n, n]}(\tau), \\ &= \int_{-1/2}^{1/2} \mathcal{S}(f') \mathcal{F}_n(f - f') df', \quad \mathcal{F}_n(f) = \frac{1}{n} \frac{\sin^2(n\pi f)}{\sin^2(\pi f)} \quad (f \neq 0).\end{aligned}$$

Idea: $\mathbb{E}[\hat{\gamma}_{\tau}] = (1 - |\tau|/n)\gamma_{\tau}$; multiplying the ACVS by the triangular weight w_{τ} in the time domain becomes convolution with $\mathcal{F}_n$ in frequency. As $n \rightarrow \infty$, $\mathcal{F}_n$ concentrates at 0, so the periodogram is *asymptotically unbiased*.

Theorem — Variance of the periodogram (inconsistency)

For $f \neq 0 \bmod \frac{1}{2}$,

$$\text{Var}(\hat{\mathcal{S}}^{(p)}(f)) \xrightarrow{n \rightarrow \infty} \mathcal{S}(f)^2 > 0,$$

and ordinates at distinct Fourier frequencies are asymptotically uncorrelated. *Idea (Gaussian case):* $J(f)$ is asymptotically complex Gaussian, so $\hat{\mathcal{S}}^{(p)}(f) = |J(f)|^2 \sim \frac{1}{2}\mathcal{S}(f)\chi_2^2$, with variance $\mathcal{S}(f)^2$. The variance **does not shrink with n** : the periodogram is *not consistent*.

Why blurring, and why the variance never dies

Observing a finite window multiplies the ACVS by a triangular weight $\rightarrow$ in frequency that is convolution with $\mathcal{F}_n$. But $\mathcal{F}_n$ has **slow-decaying side lobes**, so power *leaks* from one frequency into its neighbours and sharp peaks get smeared. That is the **bias**. Separately, more data does *not* average out the noise at a fixed f : it gives you *more* frequency ordinates, each just as noisy as before ($\approx \mathcal{S}(f) \cdot \text{Exp}(1)$). “It got noisier as I added data” is normal — finer resolution, same per-ordinate scatter. To kill variance you must *average*.

Theorem — Expectation of the tapered periodogram

With a taper $\{h_t\}$ (set $h_t = 0$ off T), the Fejér kernel is replaced by $|H|^2$, the squared modulus of the taper’s transform:

$$\mathbb{E}[\hat{\mathcal{S}}_h^{(p)}(f)] = \int_{-1/2}^{1/2} \mathcal{S}(f') |H(f - f')|^2 df'.$$

Idea: the time-domain weight is now $w_{\tau} = \sum_t h_t h_{t+\tau}$ (autocorrelation of the taper), whose transform is $|H(f)|^2$. A well-chosen taper (e.g. a dpss / Slepian taper) makes $|H|^2$ far more concentrated near 0 with much lower side lobes than $\mathcal{F}_n$ — so **tapering reduces bias/leakage**. (Corollary: if $\|h\|_2^2 = 1$, the tapered periodogram is *exactly* unbiased for white noise, $\mathbb{E}[\hat{\mathcal{S}}_h^{(p)}(f)] = \sigma^2$, by Parseval — which pins the normalisation.)

Theorem — Variance reduction by multitapering

For K orthonormal tapers the individual tapered periodograms become asymptotically uncorrelated, each with variance $\mathcal{S}(f)^2$, so averaging divides the variance by K :

$$\text{Cov}(\hat{\mathcal{S}}_{h_k}^{(p)}(f), \hat{\mathcal{S}}_{h_{k'}}^{(p)}(f)) \rightarrow \mathcal{S}(f)^2 \delta_{k,k'}, \quad \text{Var}(\hat{\mathcal{S}}^{(mt)}(f)) \approx \frac{\mathcal{S}(f)^2}{K}.$$

Idea: orthogonality of the tapers $\Rightarrow K$ near-independent estimates; averaging K uncorrelated terms divides variance by K . Letting $K \rightarrow \infty$ slowly with n restores **consistency**, at the cost of a modest increase in bias.

Intuition

The bias–variance story in one line

The raw periodogram has tolerable (vanishing) **bias** but stubborn **variance**. **Tapering** attacks bias: it swaps the leaky kernel $\mathcal{F}_n$ for a sharper $|H|^2$. **Multitapering** attacks variance: it averages K near-independent tapered estimates, so $\text{Var} \approx \mathcal{S}^2/K$. Used together they give a usable spectral estimate.

Tapering vs. truncating the ACVS

Why not just chop the ACVS to a short window to “smooth”? Because truncating a genuine ACVS generally breaks non-negative-definiteness (the resulting “SDF” can go negative). The clean way to localise is to taper the data / shape the spectrum, *not* to crop the ACVS.

Key takeaways

Key takeaways — the week’s reflexes

- **Compute it.** Periodogram $= \hat{\mathcal{S}}^{(p)}(f) = |J(f)|^2$; one FFT, always ≥ 0 , $O(n \log n)$.
- **Bias.** $\mathbb{E}[\hat{\mathcal{S}}^{(p)}] = \mathcal{S} * \mathcal{F}_n$: the Fejér kernel’s slow side lobes *blur* the SDF and leak power; sharp peaks suffer most. Asymptotically unbiased as $n \rightarrow \infty$.
- **Variance.** $\text{Var}(\hat{\mathcal{S}}^{(p)}(f)) \rightarrow \mathcal{S}(f)^2 \neq 0$: **not consistent**. More data = finer resolution, *not* less scatter.
- **Fix bias $\Rightarrow$ taper:** kernel becomes $|H|^2$ (sharper, e.g. dpss tapers); keep $\|h\|_2^2 = 1$ (unbiased for white noise).
- **Fix variance $\Rightarrow$ multitaper:** average K orthonormal tapers, $\text{Var} \approx \mathcal{S}^2/K$.